In [8]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

In [9]:
```
df = pd.read_csv("titanic_clean.csv")
```

In [10]:
```
print(df.head())
```

```
   passengerid  survived  pclass     sex   age  sibsp  parch
fare
0            1         0       3    male  22.0      1      0
7.2500
1            2         1       1  female  38.0      1      0
71.2833
2            3         1       3  female  26.0      0      0
7.9250
3            4         1       1  female  35.0      1      0
53.1000
4            5         0       3    male  35.0      0      0
8.0500
```

In [12]:
```
df.info()
```

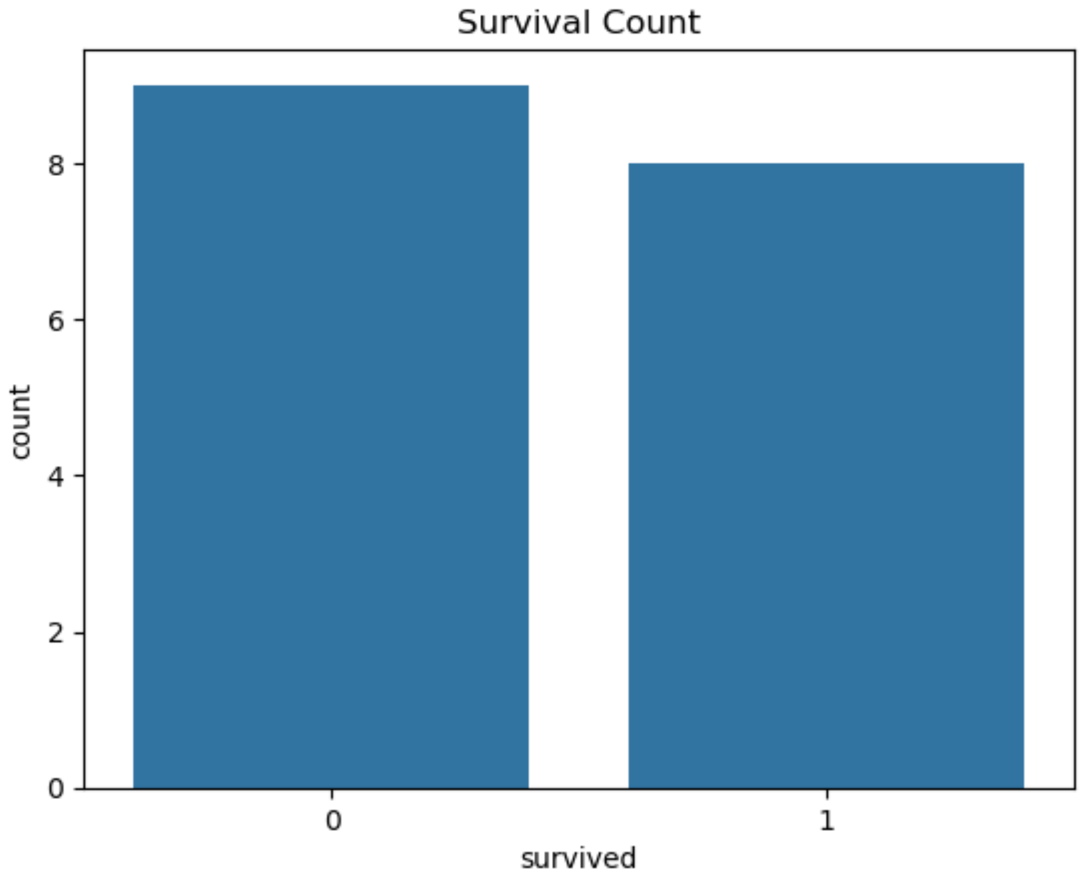
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 8 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   passengerid  20 non-null     int64
 1   survived     20 non-null     int64
 2   pclass       20 non-null     int64
 3   sex          20 non-null     object
 4   age          17 non-null     float64
 5   sibsp        20 non-null     int64
 6   parch        20 non-null     int64
 7   fare         20 non-null     float64
dtypes: float64(2), int64(5), object(1)
memory usage: 1.4+ KB
```

In [13]:
```
1 print(df.describe())
```

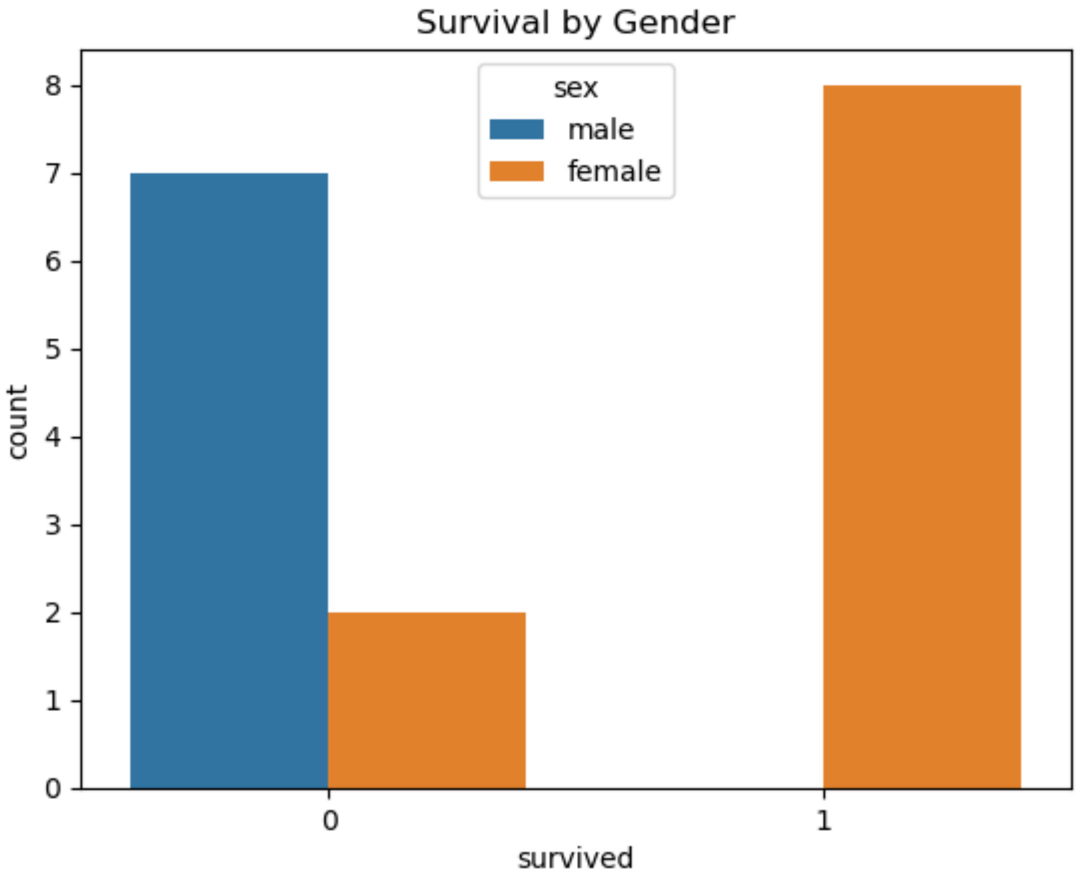
```
       passengerid   survived     pclass        age        sib
sp      parch  \
count     20.00000  20.000000  20.000000  17.000000  20.0000
00  20.000000
mean      10.50000   0.500000   2.450000  28.000000   0.7000
00   0.500000
std        5.91608   0.512989   0.825578  17.779904   1.0809
35   1.192079
min        1.00000   0.000000   1.000000   2.000000   0.0000
00   0.000000
25%        5.75000   0.000000   2.000000  14.000000   0.0000
00   0.000000
50%       10.50000   0.500000   3.000000  27.000000   0.0000
00   0.000000
75%       15.25000   1.000000   3.000000  38.000000   1.0000
00   0.250000
max       20.00000   1.000000   3.000000  58.000000   4.0000
00   5.000000

            fare
count  20.000000
mean   22.199370
std    18.058888
min     7.225000
25%     8.050000
50%    16.350000
75%    29.361450
max    71.283300
```

In [14]:
```
1 df = df.dropna()
```

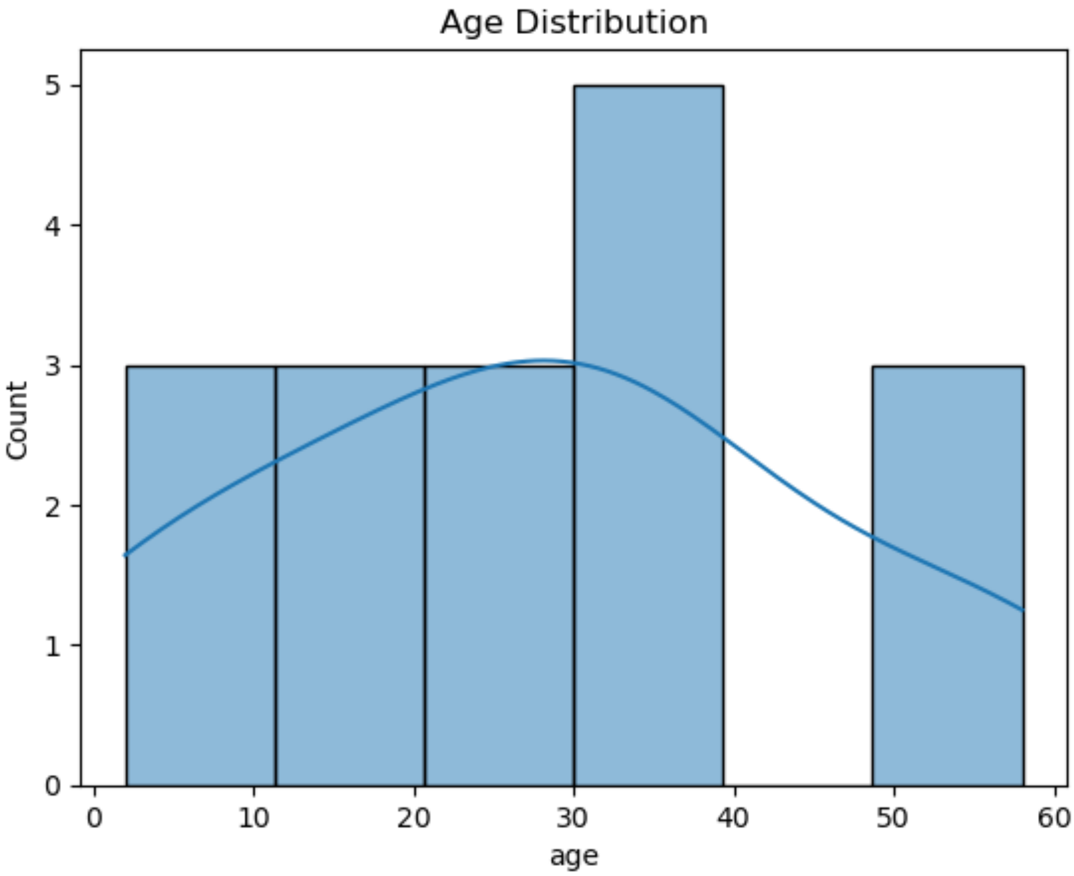
In [15]:
```
sns.countplot(x='survived', data=df)
plt.title("Survival Count")
plt.show()
```



In [16]:
```
sns.countplot(x='survived', hue='sex', data=df)
plt.title("Survival by Gender")
plt.show()
```

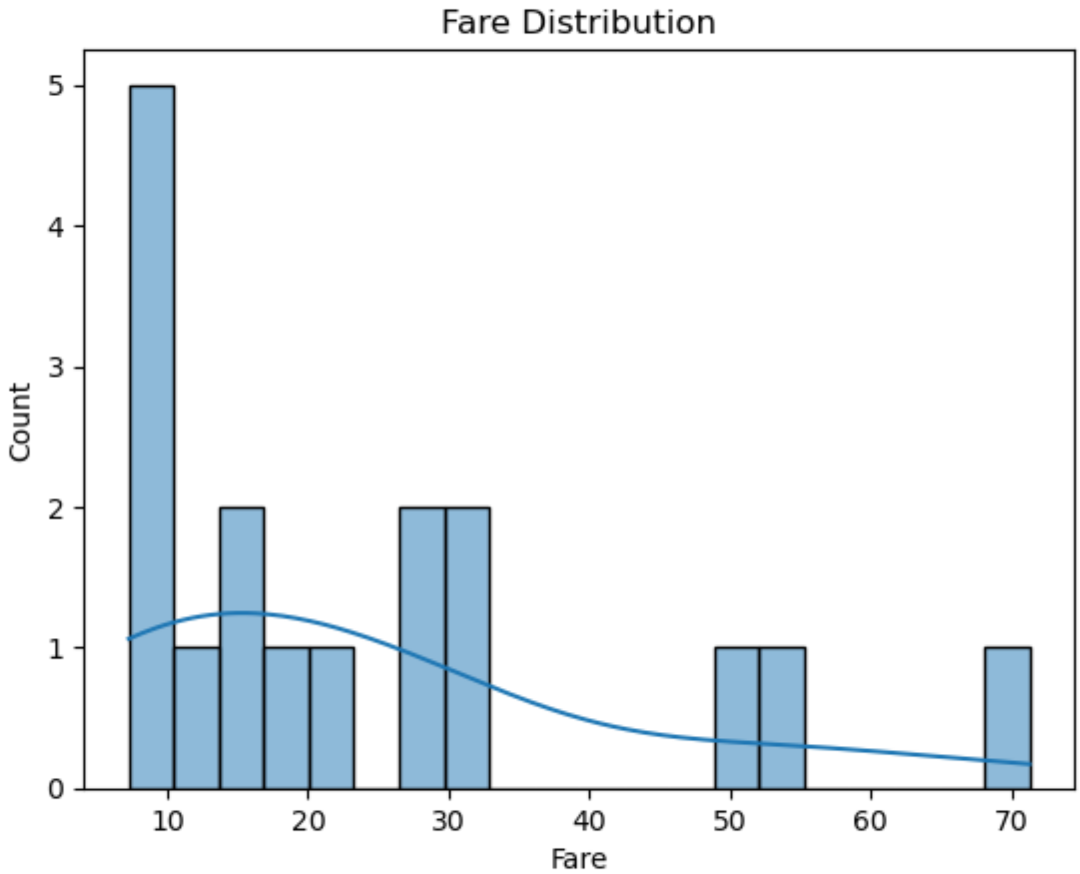


In [17]:
```
sns.histplot(df['age'], kde=True)
plt.title("Age Distribution")
plt.show()
```



In [18]:

```
sns.histplot(df['fare'], bins=20, kde=True)
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Count")
plt.show()
```



In [ ]:

```

```